Clustering

Expectation-Maximization Algorithm

Expectation-Maximization Algorithm

- Real-world applications of machine learning
  - there are many relevant features available for learning but only a small subset of them are observable.
- If the variables are observable - use the instances for the purpose of learning
- when it is not observable, then we can predict its value in the instances.

Expectation-Maximization Algorithm

- Expectation-Maximization algorithm can be used for the latent variables
- Latent variables - Variables that are not directly observable and are actually inferred from the values of the other observed variables too
- To predict their values, we can use probability distribution.

Expectation-Maximization Algorithm

This algorithm is base for many unsupervised clustering algorithms.

It was proposed in 1977 by Dempster, Nan Laird, and Donald Rubin.

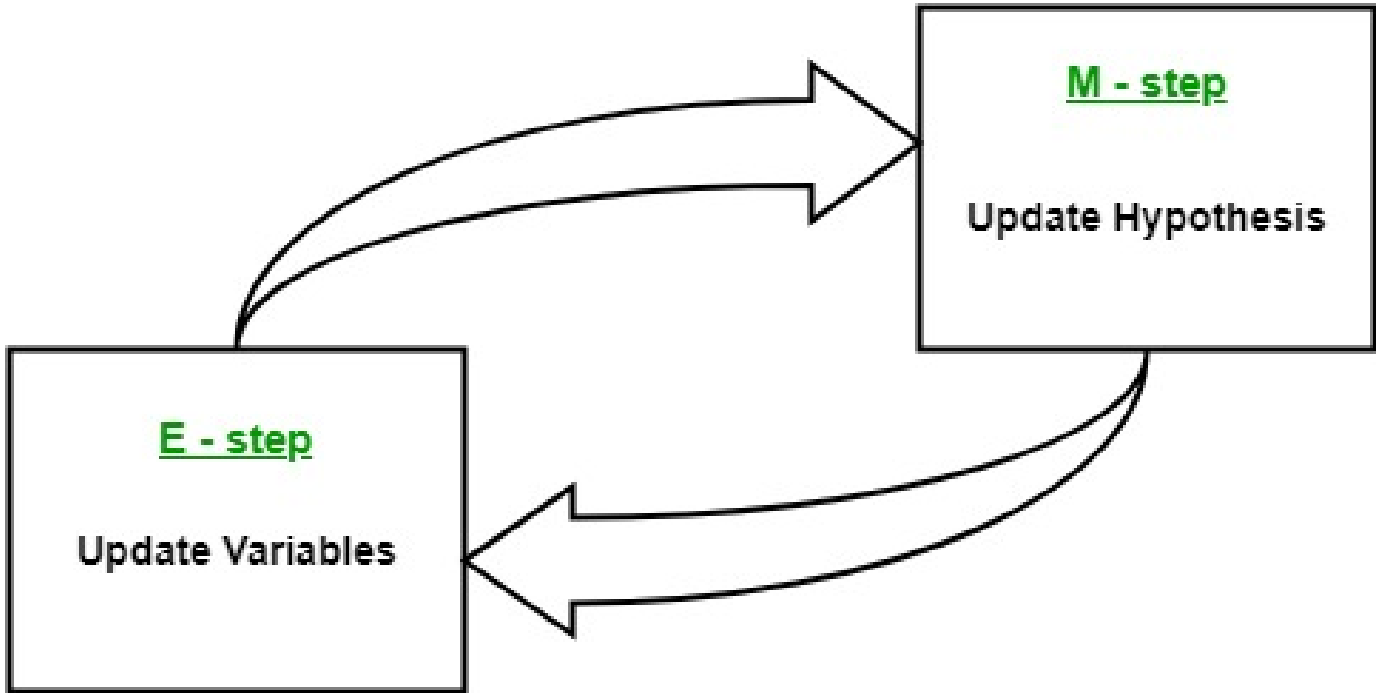
It is used to find the local maximum likelihood parameters in the cases where latent variables are involved and the data is missing or incomplete.

Algorithm

1. Given a set of incomplete data, consider a set of initial parameters.
2. Expectation step (E – step):
    1. Using the observed available data of the dataset, estimate (guess) the values of the missing data.
    2. The algorithm computes the latent variables i.e. expectation of the log-likelihood using the current parameter estimates.
3. Maximization step (M – step):
    1. Complete data generated after the expectation (E) step is used in order to update the parameters.
    2. The algorithm determines the parameters that maximize the expected log-likelihood obtained in the E step, and corresponding model parameters are updated based on the estimated latent variables.
4. Repeat step 2 and step 3 until convergence.

Algorithm

- Essence of Expectation-Maximization algorithm: use the available observed data of the dataset to estimate the missing data and then using that data to update the values of the parameters.



Important Terms used in Expectation-Maximization (EM) Algorithm

- **Latent Variables:**
  - Latent variables are unobserved variables in statistical models that can only be inferred indirectly through their effects on observable variables.
  - They cannot be directly measured but can be detected by their impact on the observable variables.

- **Likelihood:**
  - It is the probability of observing the given data given the parameters of the model.
  - In the EM algorithm, the goal is to find the parameters that maximize the likelihood.

- **Log-Likelihood:**
  - It is the logarithm of the likelihood function, which measures the goodness of fit between the observed data and the model.
  - EM algorithm seeks to maximize the log-likelihood.

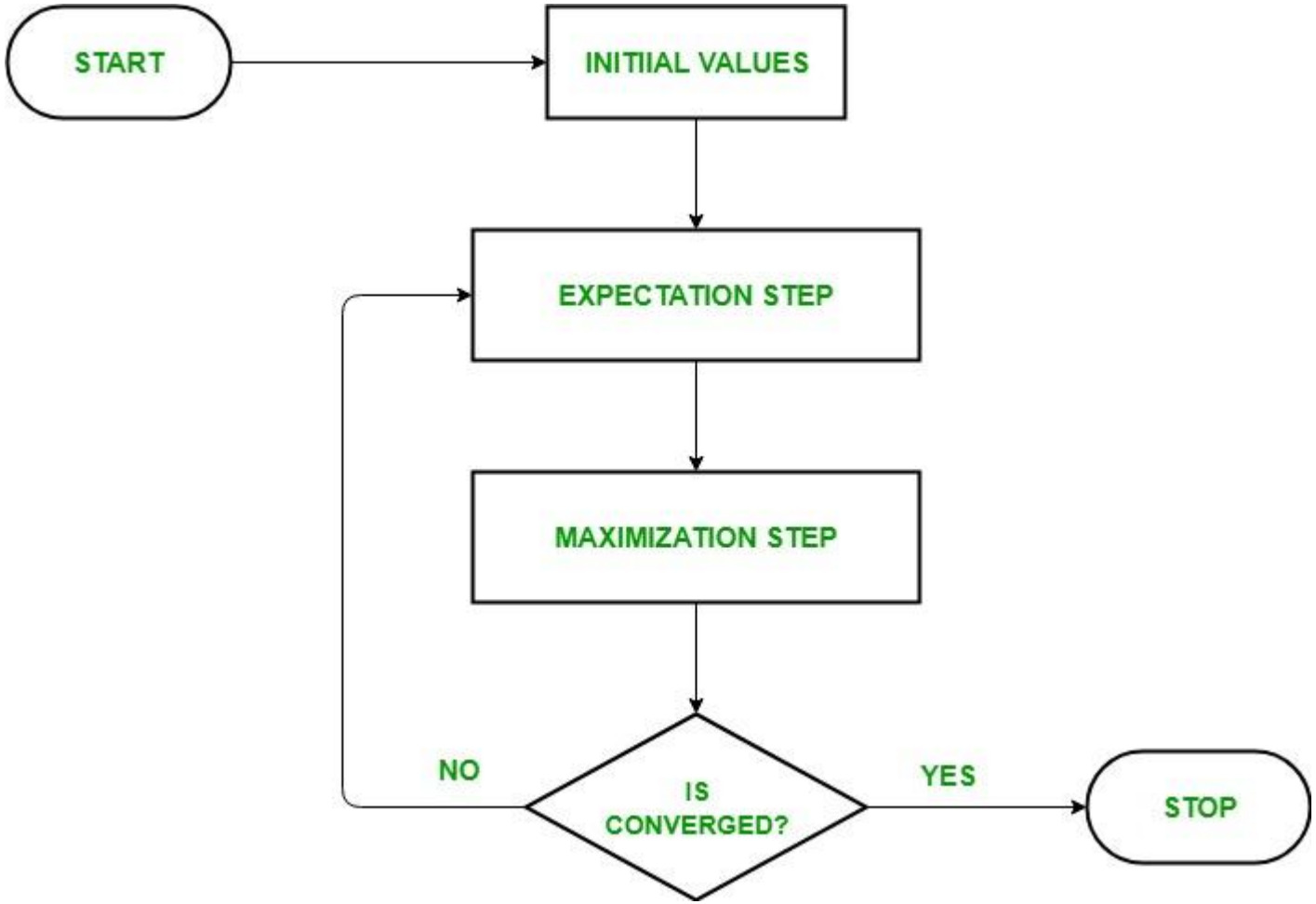
Important Terms used in Expectation-Maximization (EM) Algorithm

- **Maximum Likelihood Estimation (MLE)**:
  - MLE is a method to estimate the parameters of a statistical model by finding the parameter values that maximize the likelihood function, which measures how well the model explains the observed data.

- **Posterior Probability (MAP)**:
  - In the context of Bayesian inference, the EM algorithm can be extended to estimate the maximum a posteriori (MAP) estimates, where the posterior probability of the parameters is calculated based on the prior distribution and the likelihood function.

Expectation-Maximization (EM) Algorithm Works:

- The Expectation-Maximization algorithm use the available observed data of the dataset to estimate the missing data and then use that data to update the values of the parameters.



Usage of EM algorithm

It can be used to fill the missing data in a sample.

It can be used for the purpose of estimating the parameters of Hidden Markov Model (HMM).

It can be used for discovering the values of latent variables.

Advantages of EM algorithm

It is always guaranteed that likelihood will increase with each iteration.

The E-step and M-step are often easy for many problems in terms of implementation.

Disadvantages of EM algorithm

- It has slow convergence.
- It makes convergence to the local optima only.
- It requires both the probabilities, forward and backward.



Thank you